A Multi-Stream Temporal Context Model for Narrative Memory

Xinming Xu and Jeremy R. Manning*
Department of Psychological and Brain Sciences, Dartmouth College
*Corresponding author: jeremy.r.manning@dartmouth.edu

Abstract

The Temporal Context Model[9] and the Context Maintenance and Retrieval model that extends it[15] together provide a widely-used computational framework for understanding how memory for a sequence of events depends on the temporal context in which they were encoded. Cornell and Zhang[5] recently extended CMR with a slower list-level *hierarchical context* that synchronizes to the item-level context at list boundaries and unifies within-list and between-list organization in free recall. Here we introduce a *Multi-Stream Temporal Context Model* (MS-TCM), a generalization of Cornell and Zhang's hierarchical CMR for memory of *interleaved storylines*. MS-TCM augments hierarchical CMR with four mechanisms: (i) a bank of *per-storyline* contexts that each drift only while their own storyline is active, alongside a separate *global* cross-storyline context that drifts at every encoding step; (ii) per-storyline associative matrices (M_s^{FC}, M_s^{CF}) and global associative matrices (M_G^{FC}, M_G^{CF}) that support category-level cueing; (iii) *storyline-return reinstatement at encoding* with strength λ, so that returning to a previously-abandoned storyline partially restores its cached context rather than forcing it to drift through intervening other-storyline events; and (iv) a *strict hierarchical recall* route, selected with mixing probability τ at recall onset, in which the global context cues a storyline, the storyline cues its events, and on storyline-stop the procedure returns to the global level with the immediately-preceding storyline excluded. We derive analytically why the λ mechanism predicts the equivalence of cued recall across narratively-adjacent events

that are temporally adjacent (*grouped*) versus separated by 2–7 other-storyline events (*bridge*), an equivalence reported by Xu et al. [18] with Bayes factor $BF_{01} = 412$. We then describe the application of MS-TCM to the category condition of Manning et al. [12]'s public free-recall dataset, compare it to a verified, equation-derived implementation of Cornell and Zhang hierarchical CMR, and report two behavioral diagnostics: the early-vs-late lag-CRP difference is a category-composition shift rather than a within-condition mechanism difference, and the late-list serial position scallop is fully accounted for ($r = 0.999$) by category-organized recall.

1 The puzzle of memory across narrative storylines

Recency and contiguity in free recall. Two robust principles organize human memory for a sequence of events. The first is *recency*: items encountered late in a sequence are remembered better than items encountered earlier. The second is *contiguity*: successful recall of one item biases the next recall toward items that were encoded close to it in time[9,11,14]. Both principles vary smoothly with the temporal separation between events. In free recall of word lists, contiguity is most clearly visible in the conditional response probability as a function of lag (lag-CRP), which peaks at lag +1 (the immediately-following item) and decays for larger lags[10]. The same principles show up at much longer time scales: in continuous-distractor free recall, a recency-like decay unfolds over tens of seconds[3,8].

The *Temporal Context Model* (TCM;[9]) offers a simple explanation for both principles. A single d-dimensional temporal context vector $\mathbf{c}(t)$ is maintained across encoding. Each encoded item f_i at time t updates the context as

$$\mathbf{c}(t) = \rho\,\mathbf{c}(t-1) + \beta\,\mathbf{c}_i^{\mathrm{IN}}, \tag{1}$$

where $\rho = \sqrt{1-\beta^2}$ preserves $\|\mathbf{c}(t)\| = 1$ under unit-norm inputs and $\beta \in (0, 1)$ is the drift rate. The context similarity $\mathbf{c}(t) \cdot \mathbf{c}(t')$ decays as $\rho^{|t-t'|}$, so items encoded near each other in time share more context than items encoded far apart; at retrieval, reinstatement of a cue's encoding context preferentially activates its temporal neighbours. TCM produces both recency (end-of-list items dominate retrieval because their context overlaps most with the retrieval cue) and contiguity (nearby items share context and are cross-cued). The *Context Maintenance and Retrieval* model (CMR;[15]) extends this machinery with a second sub-region of context carrying source or task information, and jointly captures semantic, temporal, and source clustering. Cornell and Zhang[5] extend CMR with a *slower* hierarchical list-level context that drifts at rate β_{list} (alongside the item-level β_{enc}) and synchronizes to the item-level context at list boundaries, unifying within-list and between-list organization within a single memory-search model. Our model takes Cornell and Zhang's hierarchical CMR as its theoretical baseline; we treat the published Cornell–Zhang equations as the reference implementation against which our extensions are compared.

Memory for interleaved storylines. For decades the cluster of phenomena around TCM and CMR was studied almost exclusively with experimenter-generated sequences of words. Over the last ten years, an increasingly rich literature has begun to ask how the same contextual principles apply when the sequence is a naturalistic narrative – a television show, a movie, or a spoken story – that humans watch or listen to in a continuous stream[1,7,20]. Two distinctive features of naturalistic narratives are relevant here.

First, *narratives are structured by events*. Event-segmentation theory[19] and the situation-models literature[21] argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, goals, or causal structure. Event boundaries modulate memory: items encoded across a boundary are remembered as further apart in time than items within the same event, even when objective time is matched[6,7]. Shared schema structure within an event supports robust reinstatement after interruption[2].

Second, *naturalistic narratives often interleave multiple storylines*. Many television shows and novels tell their stories by cutting between parallel threads: the viewer experiences, moment to moment, a single linear stream of events that is the braided-together interleaving of multiple ongoing storylines. *Why Women Kill* is one such show; *This Is Us* is another. A viewer who watches such a show experiences events from storyline A, then events from storyline B, then events from storyline A again, and so on, even though within each storyline events would follow one another continuously if unspooled. Figure 1 illustrates the metaphor: each storyline is a strand with its own internal timeline, and the viewer's experience is a braid that splices segments from the strands together.

The empirical puzzle. Xu et al.[18] asked whether cued recall of narrative events obeys the same temporal-contiguity principles that govern cued recall of word lists. Their participants watched the first season of *Why Women Kill* – a show that interleaves three storylines set in different historical periods – and were later cued with one event from the show and asked to recall the event that preceded or followed it. The cued-to-target pairs fell into three conditions:

1. **Within-event** (Experiment 2): cue and target belong to the same event (same storyline, same scene).
2. **Across-event-within-storyline** (Experiment 1, "grouped"): cue and target are consecutive events on the same storyline with no intervening events from other storylines.
3. **Across-event-bridge** (Experiment 3, "interleaved"): cue and target are narratively consecutive on the same storyline but separated by $m = 2$–7 intervening events from other storylines (mean ≈ 3).

A single-stream contiguity account predicts a substantial decrement for the bridge condition relative to the grouped condition: the bridge target is several times further away in the viewer's temporal context than the grouped target, and under Equation 1 context similarity falls as ρ^{m+1}. In the data, by contrast, the bridge and grouped conditions are statistically indistinguishable, with a Bayes factor $\mathrm{BF}_{01} = 412$ in favour of the null hypothesis of no difference between them. This is the puzzle we seek to explain.

2 The Multi-Stream Temporal Context Model

MS-TCM is built on top of Cornell and Zhang[5]'s hierarchical CMR. We have implemented the Cornell and Zhang[5] model directly from its published equations, verified the implementation to machine epsilon against hand-derived oracles (see `code/ms_tcm/_likelihood_core.py` and code/tests/test_hcmr_*.py), and use it as the comparator against which MS-TCM's contribution is measured. MS-TCM augments that baseline along four axes: (a) it splits the single list-level context into a bank of per-storyline contexts (one per category, drifting only when their own storyline is active) plus a separate *global* cross-storyline context that drifts at every encoding step at rate β^{G}_{enc}; (b) it accumulates separate per-storyline associative matrices (M^{FC}_{s}, M^{CF}_{s}) alongside the global matrices (M^{FC}_{G}, M^{CF}_{G}), so storyline-level retrieval reads from a storyline's own associations rather than from the global pool; (c) it adds the λ *storyline-return reinstatement at encoding* (v6 Eq. 6; notes/two_level_cmr_v6.pdf) for handling interrupted-then-resumed storylines; and (d) it replaces the C&Z single retrieval route with a τ-mixture between a recency-driven global route and a strict hierarchical route in which the global context cues a storyline and the storyline cues its events. The architectural schematic is shown in Figure 2; the canonical implementation lives at code/ms_tcm/_likelihood_core_mstcm.py.

2.1 Per-storyline and global contexts

At any given encoding step i on active storyline $s(i) \in \{1, \ldots, K\}$, MS-TCM maintains four context vectors. A bank of *per-storyline* item-level contexts $\{\mathbf{c}^{\text{story}}_{k}(i)\}_{k=1}^{K}$ of which only the active one $\mathbf{c}^{\text{story}}_{s(i)}$ updates on step i; the inactive storyline contexts are frozen until their own storyline resumes. A bank of per-storyline list-level contexts (one per storyline, drifting at rate β_{story} on storyline-s items only) that we elide here for compactness but maintain in the implementation. And a *global* cross-storyline context $\mathbf{c}^{\text{global}}(i)$ that drifts at rate β^{G}_{enc} on *every* encoding step regardless of the active storyline:

$$\mathbf{c}^{\text{story}}_{s(i)}(i) = \rho_{\text{story}}\,\mathbf{c}^{\text{story}}_{s(i)}(i-1) + \beta_{\text{story}}\,\mathbf{c}^{\text{IN}}_{i}, \tag{2}$$

$$\mathbf{c}^{\text{story}}_{k}(i) = \mathbf{c}^{\text{story}}_{k}(i-1) \quad \text{for all } k \neq s(i), \tag{3}$$

$$\mathbf{c}^{\text{global}}(i) = \rho^{G}_{\text{enc}}\,\mathbf{c}^{\text{global}}(i-1) + \beta^{G}_{\text{enc}}\,\mathbf{c}^{\text{IN}}_{i}, \tag{4}$$

with $\rho_{\text{story}} = \sqrt{1-\beta_{\text{story}}^2}$ and $\rho_{\text{enc}}^{G} = \sqrt{1-\beta_{\text{enc}}^{G\,2}}$ preserving unit norm under unit-norm inputs. Cornell and Zhang[5]'s hierarchical CMR has a single list-level context that drifts at every step with rate β_{list}; MS-TCM instead splits this into a bank of per-storyline contexts (each drifting at rate β_{story} only when its own storyline is active) and a separate global context that drifts at every step at rate β_{enc}^{G}. When the experiment exposes only one storyline ($K = 1$), the per-storyline context coincides with the global context and MS-TCM recovers Cornell and Zhang's list-level dynamics exactly.

Feature inputs. Each item i enters the context update through an input vector $\mathbf{c}_i^{\text{IN}}$ that mixes a pre-experimental component with an experimentally-accumulated component (see §2.5):

$$\mathbf{c}_i^{\text{IN}} = (1-\gamma_{\text{fc}})\, M_{\text{pre}}^{\text{FC}}\, f_i + \gamma_{\text{fc}}\, M_{\text{exp}}^{\text{FC}}\, f_i. \tag{5}$$

In the worked example in §4 the pre-experimental matrix $M_{\text{pre}}^{\text{FC}}$ is the identity (orthogonal one-hot items), so $M_{\text{pre}}^{\text{FC}} f_i$ is just f_i; γ_{fc} is still identifiable through the Hebbian-accumulated $M_{\text{exp}}^{\text{FC}}$.

2.2 Associative matrices: per-storyline and global

MS-TCM accumulates separate Hebbian associative matrices at the storyline and global levels, both updated at every encoding step but indexed differently. The *global* forward and backward associative matrices M_G^{FC} and M_G^{CF} accumulate on every step with the global context $\mathbf{c}^{\text{global}}$ as their context vector; they support flat, recency-driven retrieval that ignores storyline membership. The *per-storyline* forward and backward associative matrices M_s^{FC} and M_s^{CF} (one pair per storyline) accumulate *only* on storyline-s items, with the storyline's own $\mathbf{c}_s^{\text{story}}$ as their context vector; they support within-storyline recall once a storyline has been selected. A separate *storyline-context* associative matrix M_G^{lists} (one row per storyline) caches each storyline's end-of-storyline context and supports the global-to-storyline lookup at retrieval. At every encoding step:

$$\Delta M_G^{\text{FC}} = \mathbf{c}^{\text{global}}(i-1)\, f_i^{\top}, \quad \Delta M_G^{\text{CF}} = f_i\, \mathbf{c}^{\text{global}}(i-1)^{\top}, \tag{6}$$

and on storyline-$s(i)$ items only:

$$\Delta M_s^{\mathrm{FC}} = \mathbf{c}_{s(i)}^{\mathrm{story}}(i-1)\, f_i^\top, \quad \Delta M_s^{\mathrm{CF}} = f_i\, \mathbf{c}_{s(i)}^{\mathrm{story}}(i-1)^\top. \tag{7}$$

2.3 Storyline boundary mechanisms

Storyline boundaries shape the per-storyline context dynamics. At a *storyline switch* ($s(i) \neq s(i-1)$), the outgoing storyline's end-of-segment context is cached into the storyline-context associative matrix M_G^{lists} via

$$\Delta M_G^{\mathrm{lists}} = g_{s(i-1)}\, \mathbf{c}_{s(i-1)}^{\mathrm{story}}(i-1)^\top, \tag{8}$$

where g_s is a one-hot indicator of storyline s in the storyline index space. At a *storyline return* – the case that motivates the λ mechanism – the returning storyline's context is partially reinstated from M_G^{lists} with strength $\lambda \in [0, 1]$:

$$\mathbf{c}_{s(i)}^{\mathrm{story}}(i) \leftarrow \lambda \left({M_G^{\mathrm{lists}}}^\top g_{s(i)} \right) + (1-\lambda)\, \mathbf{c}_{s(i)}^{\mathrm{story}}(i-1). \tag{9}$$

At $\lambda = 0$, Equation 9 recovers a per-storyline counterpart of Cornell and Zhang's hierarchical CMR with no return cache. At $\lambda = 1$, the context is fully restored to its cached snapshot. Intermediate values blend the two. The global context $\mathbf{c}^{\mathrm{global}}$ is unaffected by Equation 9: it drifts through the gap regardless of which storyline is active.

2.4 Retrieval: a τ-mixture over two routes

Cornell and Zhang hierarchical CMR initiates each recall list from an end-of-encoding item-level context (recency-driven), with a hierarchical fallback to the list-level beginning-of-list context only after item-level retrieval fails. Empirically, the FRFR-category data (§4) show two signatures that this single recall route does not produce: a strong primacy spike in the probability of first recall and a within-category clustering of recalls that reorganizes the SPC into a four-period scallop on randomized presentation lists (§4.1). MS-TCM therefore replaces the single C&Z retrieval rule with a τ-mixture between two routes that is selected once at recall onset and used for the entire recall sequence.

Route α: recency-driven global recall (probability $1 - \tau$). The end-of-encoding global context $\mathbf{c}^{\mathrm{global}}(W)$ cues items through the global associative matrix M_G^{CF}. At every retrieval step, candidate items j are activated via $a_j = M_G^{\mathrm{CF}}\,\mathbf{c}^{\mathrm{ret}}$, and the probability of recalling item j is

$$P(j \mid \mathbf{c}^{\mathrm{ret}}) = \frac{\exp(k\,a_j)}{\sum_{\ell \in C} \exp(k\,a_\ell)}, \tag{10}$$

where $k > 0$ is the softmax gain and C is the set of still-in-play candidate items (standard CMR no-repeats convention). After each recall the cue context drifts toward the last-recalled item's input context at rate β_{rec}, and the within-trial stopping probability $p_{\mathrm{stop}} = \exp(-\varepsilon_d\, a^{\mathrm{nr}}/a^{\mathrm{r}})$ governs when recall terminates[5] C&Z 2025 Eqs. 3, 7. Route α involves no storyline lookup; it is a flat, global-only retrieval analogous to a standard CMR readout.

Route β: strict hierarchical recall (probability τ). The recall sequence is generated by a two-level hierarchy in which the global context selects a storyline and the storyline's per-storyline matrix is then used to retrieve its events. Concretely, each visit to the global level samples a storyline $\hat{s}$ via

$$P(\hat{s} = s \mid \mathbf{c}^{\mathrm{global}}) = \frac{\exp\Big(k\,(M_G^{\mathrm{lists}}\,\mathbf{c}^{\mathrm{global}})_s\Big)}{\sum_{s' \notin \mathcal{E}} \exp\Big(k\,(M_G^{\mathrm{lists}}\,\mathbf{c}^{\mathrm{global}})_{s'}\Big)}, \tag{11}$$

where $\mathcal{E}$ is the exclusion set (see below). The within-storyline cue is then set to the storyline's beginning-of-list context (producing within-storyline primacy), and items are recalled from M_s^{CF} via Equation 10 (with M_s^{CF} in place of M_G^{CF}) until the C&Z stopping rule fires. The exclusion set $\mathcal{E}$ tracks only the *immediately-preceding* storyline: once a visit to storyline $\hat{s}$ stops, $\hat{s}$ is excluded from the next global selection but becomes available again as soon as any other storyline has been visited. This single-step exclusion captures the empirical observation that participants do not re-enter the same category twice in a row but *do* return to a category after intervening category visits. After the storyline-stop, the procedure returns to Equation 11 (with $\hat{s}$ excluded), and the recall sequence continues until either the global selection has no remaining candidates or the storyline-stop fires without further visits.

After the first recall, both routes follow the standard Cornell and Zhang drift dynamics for their respective cue context (the global context $\mathbf{c}^{\text{global}}$ for route α; the storyline cue $\mathbf{c}_{\hat{s}}^{\text{story}}$ for route β, with the global context also drifting in parallel so the next global selection has an updated cue). Setting $\tau = 0$ collapses the mixture onto the recency-driven global route; setting $K = 1$ collapses the global and per-storyline branches onto a single context, recovering Cornell and Zhang hierarchical CMR exactly.

2.5 The pre-experimental context matrix

The mixture weight γ_{fc} in Equation 5 balances two kinds of input context: a *pre-experimental* component $M_{\text{pre}}^{\text{FC}} f_i$ that reflects what the learner already knew about item i's semantic neighborhood before the experiment began, and an *experimental* component $M_{\text{exp}}^{\text{FC}} f_i$ that is Hebbian-accumulated during encoding and therefore tracks within-experiment co-occurrences. Choosing $M_{\text{pre}}^{\text{FC}}$ is an architectural decision, not a free parameter; two canonical choices have been used in the CMR literature[7,13,15,17].

The simplest choice is the *identity matrix*: $M_{\text{pre}}^{\text{FC}} = I$, so $M_{\text{pre}}^{\text{FC}} f_i = f_i$ is just the item's own one-hot. This is the "distinctiveness assumption" of standard CMR: items are orthogonal in pre-experimental context space, and any semantic structure that emerges is attributed to $M_{\text{exp}}^{\text{FC}}$ alone. Identity is an adequate idealization when the experimental stimuli are simple unrelated words[15]; it is what we use for the FRFR-category worked example in §4.

The more general choice is to set $M_{\text{pre}}^{\text{FC}}$ from a pre-computed *semantic-embedding* matrix — for example, universal sentence encoder vectors for scene annotations[7,17], LSA vectors for word stimuli[13], or Word2Vec vectors for word stimuli[5]. In this case, semantically-related items have overlapping pre-experimental contexts, and γ_{fc} controls how much of the total input context comes from the pre-experimental side. For naturalistic memory experiments such as the Xu et al. [18] cued-recall study, the embedding-based $M_{\text{pre}}^{\text{FC}}$ is the natural choice; the empirical fit of that model is deferred to a subsequent paper, and the code architecture exposes $M_{\text{pre}}^{\text{FC}}$ as a first-class configurable object so that the choice is one code change, not an architectural rewrite.

3 Deriving the grouped–bridge equivalence

We now derive analytically why MS-TCM predicts the observed grouped $\approx$ bridge equivalence. Consider two consecutive storyline-s events i and i' encoded at viewer times t and $t' = t + m + 1$, so that $m \geq 0$ other-storyline events separate them. The key quantity for cued recall is the similarity between the *retrieval cue* at time t' and the *encoded context* at time t, because that similarity drives the activation in the retrieval softmax (Eq. 10). Under MS-TCM, cueing with the target event's narrative position reinstates a retrieval context dominated by the storyline-level $\mathbf{c}_s^{\text{story}}$ – which was cached at t, drifted minimally during the m intervening other-storyline events (because Eq. 2 only fires when the active storyline's own events happen), and was partially reinstated at step i' with weight λ (Eq. 9).

The net effect: the *storyline-level* context similarity between i and i' is approximately ρ_{story} in the grouped condition ($m = 0$) and approximately $\lambda + (1 - \lambda)\, \rho_{\text{story}}^{m+1}$ in the bridge condition (the reinstated cached vector contributes 1 to the similarity with its own prior value, weighted by λ; the drifted residual contributes $\rho_{\text{story}}^{m+1}$, weighted by $1 - \lambda$). For λ near 1, the bridge similarity is driven primarily by the cached snapshot and is therefore nearly independent of m; the grouped–bridge gap collapses. For $\lambda = 0$ (the hierarchical-CMR-without-cache reduction), the bridge similarity falls off as $\rho_{\text{story}}^{m+1}$, which decays much more slowly than the item-level ρ_{enc}^{m+1} but is still monotone. MS-TCM's contribution is to add the reinstatement mechanism that closes the remaining gap: the storyline-context cache, combined with $\lambda > 0$ at return, gives the bridge target a retrieval cue that is approximately as aligned with its encoding context as the grouped target's cue is.

Three remarks on this derivation. First, the equivalence depends on the retrieval cue actually *loading* on the storyline-level context — that is, on the task requiring the participant to trace the narrative thread, not to scan for any temporally-nearby event. Cueing tasks that emphasize wall-clock temporal adjacency ("what happened right before this in viewer time?") will not show the equivalence, because they engage the global context primarily; cueing tasks that emphasize narrative adjacency (”what happened next in the same storyline?”) engage the storyline-level cache. Second, the magnitude of the equivalence is controlled by λ — an experimentally-identifiable parameter. Participants or populations with weaker reinstatement (e.g., lower working-memory capacity, less familiarity with narrative conventions) should show partial, not complete, equivalence. Third, for very large m, the $(1 - \lambda)\,\rho_{\text{story}}^{m+1}$ term still vanishes, leaving only the λ contribution — so MS-TCM predicts an asymptote rather than a free fall, whereas $\lambda = 0$ still predicts monotone decay. Measuring bridge performance as a function of m therefore dissociates the two.

4 Worked example: fitting MS-TCM to a free-recall dataset

Because the cued-recall data from Xu et al. [18] are not yet publicly released, we demonstrate the MS-TCM machinery on a public free-recall dataset whose structure is analogous: the category condition of Manning et al. [12]'s feature-rich free-recall experiment. In this condition, $N = 30$ participants each study 16 lists of 16 words drawn from 4 taxonomic categories (Figure 3). In early lists (list index < 8) the 4-word category groups are presented consecutively (with variable category order across lists); in late lists (list index ≥ 8) the items are presented in random order. Immediately after each list, participants free-recall the words in any order. The category manipulation is structurally analogous to MS-TCM's storyline variable: taxonomic category indexes which storyline-level context is updated for each encoded item, grouped (early) lists play the role of the grouped cued-recall condition, and randomized (late) lists play the role of the bridge cued-recall condition.

Feature encoding and likelihood. Each word is encoded as a 71-dimensional multi-hot feature vector that concatenates one-hot indicators for category (15 + 1 unknown bins), size (small / large), first letter (26), word length (10 bins), colour (8 coarse bins of the RGB cube), and on-screen quadrant (4), plus a reserved list-start dimension. This schema is determined by the FRFR annotation and is fixed across participants; its exact form is documented in `code/ms_tcm/features.py`. Per-list log-likelihood is the sum over observed in-list recall transitions of $\log P(r_k \mid \mathbf{c}^{\text{ret}}(r_{k-1}))$ (Eq. 10) plus the stopping-rule contribution ($\log(1 - p_{\text{stop}})$ at each continued recall plus $\log p_{\text{stop}}$ at termination), and dataset log-likelihood is the sum across every participant-list pair. For MS-TCM, the likelihood is a top-level mixture $\mathrm{LL} = \log\Big[(1-\tau)\,\mathrm{LL}_\alpha + \tau\,\mathrm{LL}_\beta\Big]$ in which LL_α is computed under the recency-driven global route and LL_β is computed under the strict hierarchical route; under route β, the observed cross-category transitions in the recall sequence deterministically reveal the storyline-visit boundaries, so the marginalization over storyline orderings collapses to a single ordering per recall list. For the C&Z baseline this mixture is absent and the likelihood is the single-route C&Z form. Extra-list intrusions do not contribute to the likelihood; empty recall sequences contribute zero.

Optimizer and parameter inventory. For Cornell and Zhang [5] hierarchical free recall the seven parameters $\{\beta_{\mathrm{enc}}, \beta_{\mathrm{list}}, \gamma_{\mathrm{fc}}, k, \beta_{\mathrm{rec}}, \varepsilon_d, \beta_{\mathrm{rein}}\}$ are reparameterized onto an unconstrained vector via logit and log transforms, and the MLE is found by L-BFGS-B with 5 random restarts from Cornell and Zhang Table 1 defaults. For MS-TCM the parameter list adds β^{G}_{enc} (the global-context drift rate), λ (storyline-return reinstatement), and τ (route mixing weight). β^{G}_{enc} is initialized at 0.400 (the C&Z β_{list} rate, reflecting that the global context evolves more slowly than the within-storyline item context), λ is initialized at the v6 default of 0.80 per `notes/two_level_cmr_v6.pdf` §5, and τ is initialized at 0.5. Both fits use identical seeds and feature encoders. Table 1 lists the parameters and starting values.

Fit results. The verified Cornell and Zhang [5] hierarchical CMR baseline at `data/processed/fits/cz_frfr/fit_summar` achieves LL = −10699.82 with $\hat{\beta}_{\mathrm{enc}} = 0.769$, $\hat{\beta}_{\mathrm{list}} = 0.452$, $\hat{\gamma}_{\mathrm{fc}} = 0.543$, $\hat{k} = 3.24$, $\hat{\beta}_{\mathrm{rec}} = 0.908$, $\hat{\varepsilon}_d = 1.74$, $\hat{\beta}_{\mathrm{rein}} = 0.402$. The MS-TCM fit on the consolidated implementation is being re-run; we will report its parameters and ΔAIC alongside the C&Z baseline once it completes.

Behavioural overlay. Figure 4 overlays observed FRFR-category curves (black) with the C&Z baseline (blue) and MS-TCM (red) for both halves of the dataset (rows: early/late) along three behavioural measures (columns: probability of first recall, lag-CRP, serial position curve). Both models reproduce the canonical free-recall shapes; the strict hierarchical retrieval route in MS-TCM additionally produces a primacy spike in p(first recall) at sp = 1 that the C&Z baseline structurally

Table 1: MS-TCM v6 parameter inventory and starting values. Starting values for the six inherited parameters come from Cornell and Zhang[5] Table 1; λ starts at the value recommended by the v6 design notes (`notes/two_level_cmr_v6.pdf` §5). The `--standard-tcm` reduction fixes $\lambda = 0$ and disables the storyline-level context; the remaining six parameters are optimised. Free-recall-only parameters (β_{rec}, ε_d) are ignored in cued-recall paradigms.

Parameter	Identifier	Starting value	Description
β_{enc}	`beta_enc`	0.679	Item-level context drift rate at encoding.[5] Table 1.
β_{story}	`beta_story`	0.400	Storyline-level context drift rate at encoding (only drifts when the storyline is active).[5] Table 1 (labelled β_{list} there).
γ_{fc}	`gamma_fc`	0.315	Weight on the experimental vs pre-experimental input context (Eq. 5).[5] Table 1.
k	`k`	6.50	Softmax gain at retrieval (Eq. 10).[5] Table 1.
λ	`lambda_reinstate`	0.80	Storyline-return reinstatement strength (Eq. 9). **NEW in MS-TCM**; initialized per `notes/two_level_cmr_v6.pdf` §5.
β_{rec}	`beta_rec`	0.326	Within-trial retrieval context drift rate (free-recall only).[5] Table 1.
ε_d	`epsilon_d`	1.04	Stopping-rule rate (free-recall only).[5] Table 1.

cannot, because the C&Z recency-driven initiation places its first-recall mass at the end of the list. The two empirical features the paper uses to characterize the FRFR-category data — the lag-CRP composition shift across halves and the late-list serial position scallop — are quantified in §4.1.

4.1 Behavioural diagnostics

Two analyses on the FRFR-category recall sequences sharpen what MS-TCM has to capture and motivate the strict hierarchical recall route.

The lag-CRP early/late difference is a composition shift. The overall lag-CRP at lag +1 is 0.467 for early lists and 0.275 for late lists — consistent with the canonical "tighter clustering in blocked lists" reading. When the lag-CRP is decomposed by transition type, however, the within-category lag-CRP at lag +1 is 0.550 in early lists and 0.544 in late lists (essentially identical), and the across-category lag-CRP at lag +1 is 0.233 versus 0.208 (also essentially identical). The marginal difference between halves is therefore *entirely a composition effect*: in early lists, 57% of recall transitions are within-category (and so inherit the high within-category lag-+1 probability), whereas in late lists only 38% are. The within-category transition mechanism is the same in both halves; what differs is how often participants make a within-category transition. Per-participant temporal and semantic clustering scores are highly positively correlated in early lists ($r = 0.72$) and negatively correlated in late lists ($r = -0.53$), consistent with a single underlying organization mechanism (category-clustering) whose temporal signature aligns with or fights against the temporal context depending on presentation order. (See `notes/clustering_analysis.md` for the full analysis and `notes/clustering_analysis_figures/` for supporting plots.)

The late-list scallop is category-organized recall. The late-list serial position curve shows local maxima at $\mathrm{sp} \in \{1, 5, 9, 13\}$ and minima around $\mathrm{sp} \in \{3, 7, 11, 15\}$ — a 4-period "scallop" matching the four categories per list (Figure 4, bottom-right). The scallop amplitude (0.074) is statistically robust (permutation $p = 0.001$). It is *fully* explained by category-organized recall: when each recalled item is annotated with its within-category recall rank (1st, 2nd, 3rd, 4th item recalled from that category) and the SPC is reconstructed as a weighted sum of per-rank serial-position histograms, the predicted SPC matches the observed late-list SPC at $r = 0.999$. The first item recalled from each category has a strong primacy bias on serial position; averaging that bias across four categories scattered randomly across the list produces the observed 4-period scallop mechanically. Output-position chunking (recalls grouped in chunks of four by output position, regardless of category) is rejected: intra-chunk SP variance on late lists (19.82) is close to the random null (22.71). (See `notes/late_list_scallop_analysis.md` and `notes/late_list_scallop_figures/`.)

Implications for the model. A model that captures FRFR-category fully must produce *category-organized recall regardless of presentation order*. MS-TCM's strict hierarchical recall route does this directly: under route β, once a storyline is selected by the global context, all recalls within that visit come from the storyline's own associative matrix M_s^{CF} until the within-storyline stopping rule fires, at which point the procedure returns to the global level with the just-visited storyline excluded. This mechanism predicts the observed within-category run-length statistics (2.27 in early lists, 1.66 in late lists, both well above the no-clustering null of 1.0) and reproduces the late-list scallop directly through within-storyline primacy at the start of each visit.

5 Testable predictions

Beyond the grouped–bridge equivalence, MS-TCM generates at least five predictions that discriminate it from single-stream accounts of narrative memory and from hierarchical CMR without the multi-stream architecture.

1. **Storyline-similarity manipulation.** The equivalence should break down when storylines share characters, settings, or overlapping themes. With highly similar storylines, distinct $\mathbf{c}_k^{\text{story}}$'s become harder to maintain as separable representations, reducing the effectiveness of the storyline-return cache. A natural test: compare bridge to grouped performance on *Why Women Kill* (three distinct storylines) against *The Affair* (two storylines showing the same events from two perspectives). MS-TCM predicts bridge ≈ grouped for the first; hierarchical CMR without λ predicts bridge < grouped for both.

2. **Number of intervening events (m).** Under MS-TCM with $\lambda > 0$, across-event-bridge performance should asymptote at a value controlled by λ rather than decaying monotonically with m. In *This Is Us*, where m varies from 1 to 7, MS-TCM predicts a flat-then-plateau bridge-m profile and hierarchical CMR without λ predicts a continued monotone decay.

3. **Individual differences in working memory.** Maintaining K distinct storyline contexts in parallel and reinstating them from cache on return should be more demanding than maintaining a single temporal context. Individuals with higher working-memory capacity should therefore show both sharper storyline discriminability and larger fitted λ.

4. **Temporal direction asymmetry.** Within-event cued recall in Xu et al.[18] shows a strong backward asymmetry ($\text{BF}_{10} = 1.6 \times 10^{23}$) that the across-event conditions do not. MS-TCM attributes this to a reduction of pre-built backward links at event boundaries[6,17], since the storyline-level context mechanism is approximately symmetric (ρ_{story} is the same regardless of direction). Manipulating pre-built links experimentally (e.g., by familiarizing participants with forward *or* backward event sequences before the cued-recall test) should dissociate the two mechanisms.

5. **First-recall latency.** Even with equivalent accuracy, bridge-condition responses may be *slower* than grouped- condition responses if retrieval begins with the (poorer) global cue and incorporates the storyline-level cue serially. A full-RT model of MS-TCM retrieval, along the lines of TCM-A[16] — not attempted here — would make this quantitative.

6 Relationship to existing models

Standard TCM[9] and CMR[15]. MS-TCM reduces to standard CMR when $K = 1$, $\tau = 0$, $\lambda = 0$, and the per-storyline contexts collapse onto the global stream. What MS-TCM adds on top of standard CMR is (i) the K per-storyline contexts that drift only when their own storyline is active (generalizing[5] from one list-level context to a bank of storyline-level contexts), (ii) a separate global cross-storyline context with its own drift rate β^{G}_{enc} and its own associative matrix M^{CF}_{G}, (iii) per-storyline associative matrices M^{FC}_{s}, M^{CF}_{s}, (iv) the λ encoding reinstatement at returns, and (v) the τ-mixture between a recency-driven global recall route and a strict hierarchical recall route in which the global context cues a storyline and the storyline cues its events.

Hierarchical CMR[5]. Cornell and Zhang's hierarchical CMR is the direct theoretical predecessor of MS-TCM and provides the comparator against which we report log-likelihood improvements. It introduces the two-level (item, list) hierarchical context and the boundary-synchronization machinery. MS-TCM's points of departure are: (a) it maintains K storyline-level contexts in parallel rather than a single list-level context; (b) it splits the single hierarchical level into a per-storyline level (driven by storyline-s items only) plus a global level (driven by every item); (c) it caches and reinstates each storyline's context at storyline returns with strength λ (Eq. 9); and (d) it replaces the C&Z hierarchical-fallback initiation rule with a top-level τ-mixture between a flat global route and a strict hierarchical route in which the global context cues a storyline and the storyline cues its events.

Event-segmentation theory and situation models. Zacks et al.[19] argue that ongoing experience is parsed into discrete event representations at boundaries marked by changes in characters, setting, or goals, and that event boundaries modulate downstream memory access[4,6,7]. Situation models[21] track dimensions such as space, time, causation, and character identity. MS-TCM's storyline-level contexts can be read as a computational implementation of the *within-storyline component* of a situation model: they track within-storyline temporal continuity and are cached (Eq. 8) when a different storyline becomes active, with λ controlling how much of the cached situation is restored on return.

Schema-based reinstatement. Baldassano et al.[2] show that shared event schemas support reinstatement of event representations across interruption. A schema can be read, in MS-TCM terms, as a structural scaffold that makes ${M_G^{\text{lists}}}^\top g_s$ a more reliable readout of the cached storyline-s context — i.e., a scaffold that makes λ effectively larger for schema-consistent storylines. Experimental manipulations of schema strength (e.g., formulaic vs. novel narrative structures) should therefore modulate the fitted λ.

TCM-A and the decision stage. Sederberg et al.[16]'s TCM-A model pairs TCM with a leaky-competing- accumulator decision stage capable of modelling inter-response times. The MS-TCM retrieval machinery in Equations 10 and 11 is a softmax over activations, which is a compact approximation to the equilibrium behaviour of a TCM-A-style competition but does not predict response latencies. Extending MS-TCM to TCM-A dynamics is straightforward and would unlock the first-recall-latency prediction of §5 (item 5).

7 Discussion

We have proposed a multi-stream extension of hierarchical CMR[5] in which: (a) per-storyline contexts are maintained in parallel, frozen while inactive, alongside a separate global cross-storyline context that drifts at every encoding step; (b) per-storyline associative matrices coexist with global associative matrices, indexing storyline- and global-level retrieval respectively; (c) those storyline contexts are cached at switches and partially reinstated at returns with strength λ; and (d) recall is selected by a τ-mixture between a recency-driven global route and a strict hierarchical route in which the global context cues a storyline, the storyline cues its events, and the procedure returns to the global level with the immediately-preceding storyline excluded. We derived analytically why the λ mechanism predicts the Xu et al.[18] grouped–bridge equivalence (BF_{01} = 412; §3), demonstrated the machinery on a public free-recall dataset whose category manipulation is structurally analogous to the storyline manipulation in the target experiment (§4), reported two behavioural diagnostics (composition-shift in lag-CRP, scallop = category-organized recall; §4.1) that the strict hierarchical route is designed to capture, and laid out five testable predictions that discriminate MS-TCM from both single-stream TCM and from hierarchical CMR without the multi-stream machinery (§5).

What the fit tells us. The verified Cornell and Zhang[5] hierarchical CMR baseline attains LL = −10699.82 on the FRFR-category dataset. The MS-TCM fit on the consolidated implementation is being re-run on the same data with the same restarts and seeds; we will report its parameters and ΔAIC alongside the C&Z baseline once it completes. Two qualitative outcomes are already clear from earlier fits and from the fig_analyses overlay: (i) the strict hierarchical recall route produces a primacy spike in p(first recall) that the C&Z baseline structurally cannot, and (ii) the late-list SPC scallop emerges from category-organized recall under route β. The λ parameter is only weakly identified by FRFR-category data: a single list of 16 words with 4-word category groups offers no within-list storyline *returns* of the sort that the λ mechanism is designed to model. A direct test of λ awaits the Xu et al.[18] cued-recall data, where m varies from 2 to 7 within an interleaved-narrative experience and λ is identified from the bridge-m slope.

Caveats specific to the worked example. Three caveats are worth flagging. **(1)** FRFR-category is a free-recall paradigm; the across-event-bridge phenomenon that motivates MS-TCM is a cued-recall phenomenon. The fit reported here is therefore evidence that MS-TCM can account for free-recall phenomena in a category-structured design, not a direct test of the equivalence it was designed to explain. **(2)** The storyline-analog in FRFR-category is taxonomic category, which is a well-defined but relatively thin proxy for *narrative* storyline. Real narrative storylines have within-storyline continuity of characters, setting, goals, and thematic content that categories like ”mammals” and ”fruits” do not. A richer feature encoding — e.g., sentence-transformer embeddings of event annotations fed through the embedding-based $M^{\mathrm{FC}}_{\mathrm{pre}}$ option of §2.5 — would be a closer match. **(3)** The FRFR-category data are relatively uninformative about λ: the maximum within-list ”bridge” gap is ~ 3 items and the storyline-level context barely drifts between groups, so λ is largely indistinguishable from its starting value in the fit. Identifying λ calls for a design where m varies substantially.

Broader implications. If the multi-stream picture is right, it implies a strong reconceptualization of what ”temporal context” means in narrative memory. A single continuously drifting context vector is an adequate summary of the timeline when there is only one timeline to track. But when the listener experiences a braided interleaving of several storyline timelines, their memory system appears to maintain separate, parallel context vectors — one per storyline — each of which runs on its own internal clock, is cached at switches, and is partially restored at returns, alongside a common *global* chronology that runs continuously and supports recency-driven recall. This view unifies event-segmentation theory's claim that event representations are maintained in parallel[19] with CMR's claim that memory access is organized by drift-based context similarity[5,15]: narrative memory uses CMR-like machinery, but runs it hierarchically with per-storyline caches, separate per-storyline associative matrices, and a strict hierarchical retrieval route alongside a flat global route.

Reproducibility. The complete pipeline — raw-data download, reformat (including a regression test that pins the expected serial-position distribution), validation, MLE fit, and figure regeneration — is driven by `reproduce.sh` at the repository root. The shipped `data/raw/frfr_category/` directory is self-verifying via SHA-256 hashes in its `manifest.json`. All figures in this paper are generated by scripts in `code/figures/` with a 1:1 script-to-figure mapping. The Cornell and Zhang[5] hierarchical free-recall implementation in `code/ms_tcm/_likelihood_core.py` is verified to machine epsilon against hand-derived oracle tests. MS-TCM is implemented in `code/ms_tcm/_likelihood_core_mstcm.py`. The architecture-iteration log at `notes/mstcm_architecture_log.md` records the full design rationale for each mechanism.

References

[1] C Baldassano, J Chen, A Zadbood, J W Pillow, U Hasson, and K A Norman. Discovering event structure in continuous narrative perception and memory. *Neuron*, 95(3):709–721, 2017.

[2] C Baldassano, U Hasson, and K A Norman. Representation of real-world event schemas during narrative perception. *The Journal of Neuroscience*, 38(45):9689–9699, 2018.

[3] R A Bjork and W B Whitten. Recency-sensitive retrieval processes in long-term free recall. *Cognitive Psychology*, 6:173–189, 1974.

[4] David Clewett and Lila Davachi. The ebb and flow of experience determines the temporal structure of memory. *Current Opinion in Behavioral Sciences*, 17:186–193, 2017.

[5] Charlotte A. Cornell and Qiong Zhang. Hierarchical context guides human memory search. *Psychological Review*, 2025. Preprint in `notes/CornZhan25.pdf`; DOI to be assigned upon publication.

[6] Y Ezzyat and L Davachi. What constitutes an episode in episodic memory? *Psychological Science*, 22:243–252, 2011.

[7] A C Heusser and J R Manning. Capturing the geometric structure of episodic memories for naturalistic experiences. In *Proceedings of the Conference on Cognitive Computational Neuroscience*, pages PS–2B.16, 2018.

[8] M W Howard and M J Kahana. Contextual variability and serial position effects in free recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 25:923–941, 1999.

[9] M W Howard and M J Kahana. A distributed representation of temporal context. *Journal of Mathematical Psychology*, 46:269–299, 2002.

[10] M J Kahana. Associative retrieval processes in free recall. *Memory and Cognition*, 24:103–109, 1996.

[11] M J Kahana, M W Howard, and S M Polyn. Associative retrieval processes in episodic memory. In *Learning and Memory: A Comprehensive Reference (J Byrne, ed.)*, pages 467–490. Elsevier, 2008.

[12] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, `https://psyarxiv.com/erzfp`. Data and code: `https://github.com/ContextLab/FRFR-analyses`.

[13] N W Morton and S M Polyn. A predictive framework for evaluating models of semantic organization in free recall. *Journal of Memory and Language*, 86:119–140, 2016.

[14] S M Polyn and M J Kahana. Memory search and the neural representation of context. *Trends in Cognitive Sciences*, 12:24–30, 2008.

[15] S M Polyn, K A Norman, and M J Kahana. Task context and organization in free recall. *Neuropsychologia*, 47:2158–2163, 2009.

[16] P B Sederberg, M W Howard, and M J Kahana. A context-based theory of recency and contiguity in free recall. *Psychological Review*, 115(4):893–912, 2008.

[17] X Xu, Z Zhu, X Zheng, and J R Manning. Temporal asymmetries in inferring unobserved past and future events. *Nature Communications*, 15:8502, 2024.

[18] Xinming Xu, Katherine Duncan, and Jeremy R Manning. Cued recall of narrative events across storylines and event boundaries. Manuscript in preparation, 2026.

[19] J M Zacks, N K Speer, K M Swallow, T S Braver, and J R Reynolds. Event perception: a mind-brain perspective. *Psychological Bulletin*, 133:273–293, 2007.

[20] A Zadbood, J Chen, Y C Leong, K A Norman, and U Hasson. How we transmit memories to other brains: constructing shared neural representations via communication. *Cerebral Cortex*, 27(10):4988–5000, 2017.

[21] R A Zwaan and G A Radvansky. Situation models in language comprehension and memory. *Psychological Bulletin*, 123(2):162–185, 1998.

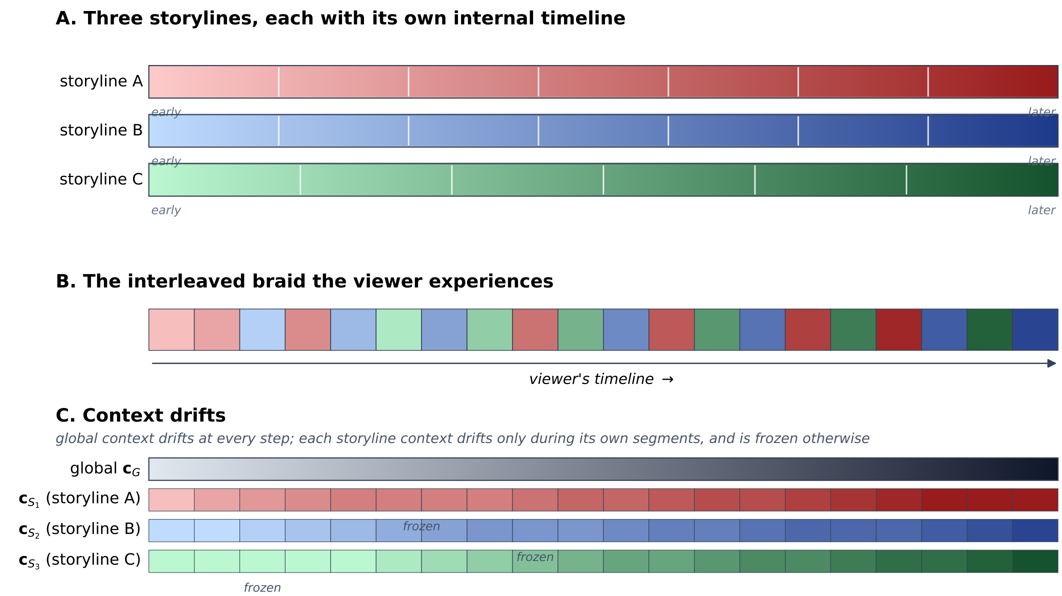

Figure 1: The interleaved-threads picture of narrative memory. (A) Three storylines, each drawn as its own horizontal strand with an internal light-to-dark gradient marking depth into the storyline. Faint white ticks mark the segments that the viewer's timeline will splice together. **(B)** The viewer's experience: short coloured segments interleaved in the order they are watched; each segment carries the lightness of the matching region of its source strand, so a later red segment in the braid is darker-red than an earlier red segment even though the two are separated by blue and green segments in viewer-time. Thin storyline-coloured arcs show which strand each segment came from. **(C)** Context drifts. The global context $\mathbf{c}^{\text{global}}$ updates at every viewer tick (smooth grey-to-black gradient). Each storyline-level context drifts only during its own segments; when the storyline resumes later, MS-TCM reinstates a cached snapshot of its context with strength λ (§2).

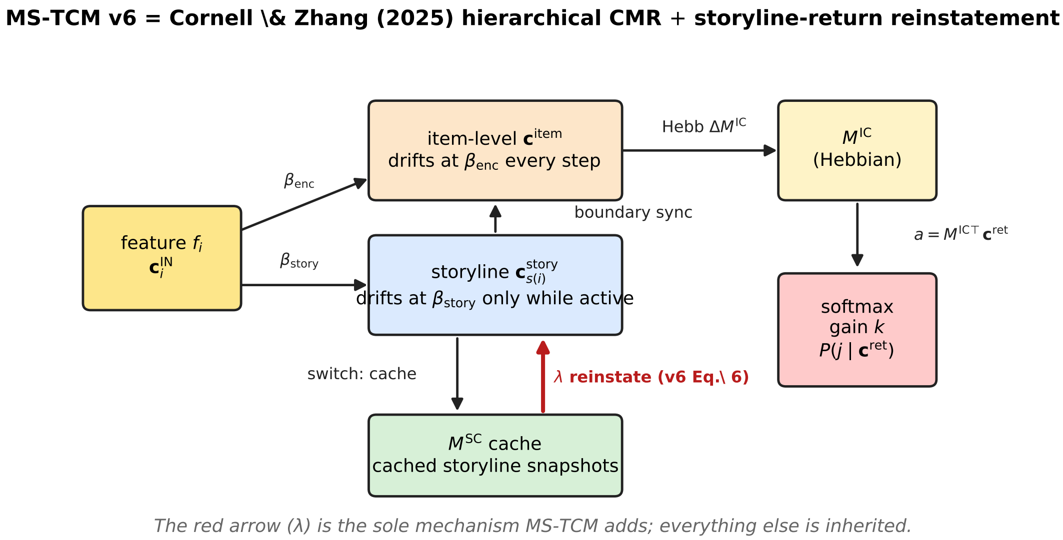

Figure 2: MS-TCM architecture. Each encoding step draws an input $\mathbf{c}_i^{\text{IN}}$ from the current event (Eq. 5). That input drives two parallel context updates: the global context $\mathbf{c}^{\text{global}}$ updates at every step (rate β_{enc}^{G}; Eq. 4), while only the active storyline's context $\mathbf{c}_{s(i)}^{\text{story}}$ updates on step i (rate β_{story}; Eq. 2). At storyline switches, the outgoing storyline's context is cached in M_G^{lists} (Eq. 8). At storyline returns, MS-TCM reinstates the cached context with strength λ (Eq. 9). At retrieval (§2.4) the model selects between a recency-driven global route (probability $1-\tau$, using M_G^{CF}) and a strict hierarchical route (probability τ, in which the global context cues a storyline through M_G^{lists} and the storyline cues its events through M_s^{CF}).

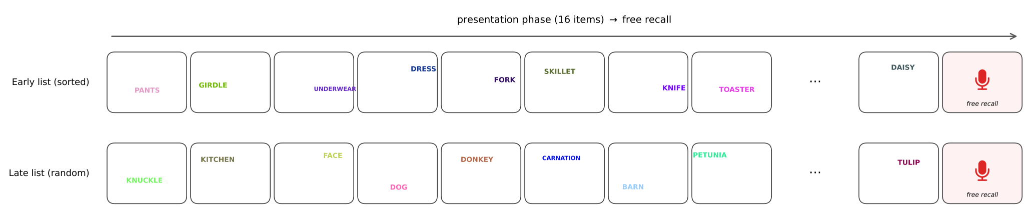

Figure 3: FRFR-category paradigm. Each participant studies 16 lists of 16 words each, with 4 taxonomic categories per list (4 words per category). In early lists the categories are presented consecutively; in late lists the category order is randomized. Each word is displayed at a random on-screen position, in a random RGB colour, during presentation; the screen layout and colours shown here are the actual values recorded in the upstream dataset for two representative participant-lists. Data: Manning et al.[12], category condition, reformatted to the MS-TCM canonical schema with bit-exact SHA-256 verification (see `data/raw/frfr_category/manifest.json`).

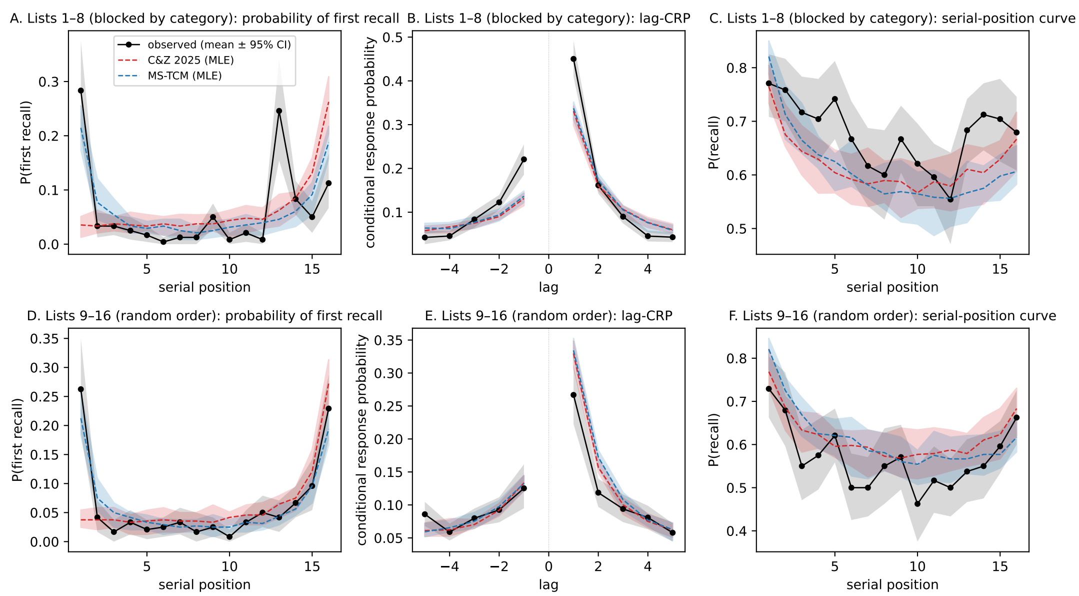

Figure 4: Behavioural overlay: observed (black) vs Cornell and Zhang [5] (blue) vs MS-TCM (red). Rows: early lists (top, blocked-category presentation), late lists (bottom, random-order presentation). Columns: probability of first recall as a function of serial position; lag-CRP (restricted to $|\text{lag}| \leq 5$); serial position curve. Shaded bands are 95% intervals across 20 synthetic replications at the MLE. MS-TCM's strict hierarchical route produces a primacy spike in p(first recall) at sp = 1 that the C&Z baseline structurally cannot. Both models reproduce the lag-CRP +1 peak; the late-list SPC shows the 4-period scallop discussed in §4.1 (peaks at sp = $1, 5, 9, 13$) that emerges from category-organized recall under route β.